# HEARTCARE AI

Smart Cardiovascular Disease Prediction & Risk Assessment Report

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| **Report ID:** HC-RPT-20260819-0018 | **Assessment Date:** 19 Aug 2026, 20:07 |
| **User Identifier:** prathm | **Model Engine:** XGBoost v1.0 + SHAP |

## AI Risk Assessment Result

**Calculated CVD Risk Score:** **83.7%** **Risk Probability:** 0.8371 **Risk Category:** HIGH

## Confirmed Health Parameters

| Parameter | Recorded Value | Clinical Range / Status |
|---|---|---|
| Age | 54.0 years | Adult |
| Gender | Male | - |
| Height / Weight | 172.0 cm / 84.0 kg | BMI: 0.0 kg/m² |
| Blood Pressure | 145.0 / 92.0 mmHg | Systolic / Diastolic |
| Cholesterol | Above Normal (200-239 mg/dL) | Total Cholesterol Level |
| Glucose | Normal (<100 mg/dL) | Fasting Glucose Level |
| Smoking Habit | Yes | Lifestyle Factor |
| Alcohol Intake | No | Lifestyle Factor |
| Physical Activity | Yes | Regular Activity |

## SHAP Model Risk Drivers (Feature Contribution Breakdown)

*Note: SHAP values quantify how much each specific feature increased or decreased the XGBoost model's risk score for this prediction. SHAP reflects model logic, not direct medical causation.*

| | | |
|---|---|---|
| ap_hi | +0.0000 | Increases Model Risk |
| cholesterol | +0.0000 | Increases Model Risk |
| ap_lo | +0.0000 | Increases Model Risk |
| gluc | +0.0000 | Increases Model Risk |
| weight | +0.0000 | Increases Model Risk |

## Personalized Non-Pharmacological Guidance

| Category & Priority | Recommendation Guidance |
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|---|---|
| **Blood Pressure Monitoring** [High Priority] | **Regular Blood Pressure Monitoring & Care** Your recorded blood pressure (145/92 mmHg) is elevated. We recommend tracking your morning and evening blood pressure at home and sharing persistent elevated readings with your physician. Reducing dietary sodium intake to under 2,000 mg/day can also support healthy blood pressure levels. |
| **Cholesterol** [High Priority] | **Heart-Healthy Nutrition & Lipid Management** Your cholesterol level is above the optimal reference range. Adopt a diet low in saturated fats and trans-fats while increasing soluble fiber intake (oats, legumes, fruits, vegetables). A complete fasting lipid panel (HDL, LDL, Triglycerides) is recommended for clinical evaluation. |
| **Professional Medical Follow-up** [High Priority] | **Physician Consultation & Medical Evaluation** Based on your calculated cardiovascular risk level (HIGH), we strongly advise scheduling a comprehensive cardiovascular consultation with a physician or cardiologist. Bring your health summary report and SHAP risk driver breakdown to your appointment. |
| **Smoking** [Moderate Priority] | **Tobacco & Smoking Cessation Support** Tobacco smoke contains compounds that impair vascular function and elevate heart disease risk. Complete smoking cessation is one of the most effective steps to improve cardiovascular health. Consider speaking with a doctor about evidence-based behavioral support and quitting programs. |
| **Weight Management** [Moderate Priority] | **Healthy Weight & Body Composition Guidance** Your calculated Body Mass Index (BMI: 28.4 kg/m²) indicates weight above the normal range. A gradual, sustainable weight reduction of 5–10% through balanced nutrition and routine activity significantly improves blood pressure and metabolic health. |
| **Diet** [General Priority] | **Cardioprotective Dietary Pattern** Emphasize a Mediterranean or DASH-style diet rich in whole grains, vegetables, lean proteins, nuts, and healthy fats (such as olive oil). Minimize processed foods and sodium. |
| **General Lifestyle** [General Priority] | **Stress Management & Quality Sleep** Ensure 7–9 hours of restorative sleep per night and practice stress reduction techniques such as mindfulness, deep breathing, or structured relaxation. |

**IMPORTANT MEDICAL DISCLAIMER:**
**This report provides an AI-assisted cardiovascular risk assessment for educational and decision-support purposes. It is not a definitive medical diagnosis and does not replace evaluation by a qualified healthcare professional. Recommendations are educational lifestyle/monitoring guidance and are NOT medical prescriptions.**